

Yeonsu Chang

Integrated Ph.D Candidate

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Graph Theory

Structural Graph Theory

Parameterized Complexity

Reduced Parameter

Research Statement

My research focuses on structural aspects of graphs and related width parameters, with emphasis on characterizations and algorithmic consequences in parameterized complexity.

Education

Integrated Ph.D in Mathematics

Mar 2022 – Present

Hanyang University, Seoul, South Korea

Advisor: O-joung Kwon

B.S. in Mathematics

Mar 2015 – Feb 2022

Hanyang University, Seoul, South Korea (incl. two years of military service)

Military Service

Republic of Korea Army

Jan 2017 – Oct 2018

Completed mandatory military service (Sergeant)

Journal Papers

3. **Pre-assignment problem for unique minimum vertex cover on bounded clique-width graphs**
Shinwoo An, Yeonsu Chang, Kyungjin Cho, O-joung Kwon, Myounghwan Lee, Eunjin Oh, Hyeonjun Shin.
Theoretical Computer Science 1084 (Sep, 2026), 116173, *AAAI2025 accepted*, [arXiv:2408.09591](https://arxiv.org/abs/2408.09591)
2. **A new width parameter of graphs based on edge cuts: α -edge-crossing width**
Yeonsu Chang, O-joung Kwon, and Myounghwan Lee.
Discrete Applied Mathematics 380 (Feb, 2026), *WG23 accepted*, [arXiv:2302.04624](https://arxiv.org/abs/2302.04624)
1. **A characterization of graphs of radius- r flip-width at most 2**
Yeonsu Chang, Sejin Ko, O-joung Kwon, and Myounghwan Lee.
Discrete Mathematics 348(4) (Apr, 2025), 114366, [arXiv:2306.15206](https://arxiv.org/abs/2306.15206)

Refereed Conference Papers without Journal Version

1. **Moderately beyond clique-width: reduced component max-leaf and related parameters**
Édouard Bonnet, Yeonsu Chang, Julien Duron, Colin Geniet, and O-joung Kwon.
ESA 2026 accepted, [arXiv.2604.19138](https://arxiv.org/abs/2604.19138)

Preprints

2. **Max Independent Set Remains NP-hard when Excluding a Planar Induced Minor**
Édouard Bonnet and Yeonsu Chang.
[arXiv.2609.11285](https://arxiv.org/abs/2609.11285)

1. **The optimal chromatic bound for even-hole-free graphs without induced seven-vertex paths**
Yeonsu Chang, Shenwei Huang, and Yidong Zhou.
[arXiv:2602.04403](#)

Talks

MC2 Seminar at ENS de Lyon	Sep 24, 2026
<i>Reduced Component Max-Leaf: Applications and Further Results</i>	
European Symposium on Algorithms 2026 (ESA 2026)	Aug 31– Sep 2, 2026
<i>Moderately beyond clique-width: reduced component max-leaf and related parameters</i>	
ASIACOMB 2026	Aug 24– 28, 2026
<i>Algorithmic Applications of Reduced component max-leaf</i>	
Korean Mathematical Society 80th Anniversary Conference	Jun 22– 25, 2026
<i>The optimal chromatic bound for (even-hole, P_7)-free graphs</i>	
Korean Student Combinatorics Workshop 2026 Winter (KSCW2026W)	Feb 2– 6, 2026
<i>Structural and Algorithmic properties of Reduced Component Max-Leaf</i>	
2025 Korean Mathematical Society Annual Meeting	Oct 22– 24, 2025
<i>Structural and Algorithmic properties of Reduced Component Max-Leaf</i>	
Math Colloquium at Hanyang University	Nov 4, 2024
<i>Combinatorial game and graph structure: cops and robber game and flipper game</i>	
2023 KMS Annual Meeting	Oct 26– 28, 2023
<i>A characterization of graphs of radius-r flip-width at most 2</i>	

Teaching Experiences

Linear Algebra 1, Hanyang University	Spring 2025
Math Capstone PBL (Solving Graph Problems), Hanyang University	Fall 2024
Linear Algebra 1, Differential Geometry, Hanyang University	Spring 2024
Head TA, Hanyang University	Fall 2022 – Spring 2024
Math Capstone PBL (Applications of graph algorithms), Hanyang University	Fall 2022
Linear Algebra 1, Hanyang University	Spring 2022

Workshops and Conferences (In Korea)

2026 Summer School on Combinatorics and Algorithms	Aug 10– 14, 2026
KAIST, Daejeon, South Korea	
2026 Korean Student Combinatorics Workshop (KSCW 2026)	Jul 27– 31, 2026
Gongju, South Korea	
37th KIAS Combinatorics workshop	May 28– 30, 2026
Jeju, South Korea	
36th KIAS Combinatorics workshop	Mar 13– 14, 2026
KIAS, Seoul, South Korea	

2025 Korean Student Combinatorics Workshop Winter (KSCW 2025W) Yeosu, South Korea	Feb 2– 6, 2026
Workshop on Algorithm for Student 2026 POSTECH, Pohang, South Korea	Jan 26– 28, 2026
35th KIAS Combinatorics workshop Busan, South Korea	Dec 22– 24, 2025
2025 KMS Fall Meeting Seoul, South Korea	Oct 22– 24, 2025
2025 Korean Student Combinatorics Workshop (KSCW 2025) Gyeongju, South Korea	Aug 20– 24, 2025
2025 Combinatorics Workshop IBS, Daejeon, South Korea	Aug 18– 20, 2025
2025 Summer School on Combinatorics and Algorithms POSTECH, Pohang, South Korea	Jul 14– 17, 2025
34th KIAS Combinatorics workshop Jeju, South Korea	May 29– 31, 2025
2025 KMS Spring Meeting KAIST, Daejeon, South Korea	Apr 24– 26, 2025
IBS-DIMAG Winter School on Graph Minors IBS, Deajeon, South Korea	Feb 3– 19, 2025
One Day Workshop on Graph Theory Ajou University, Suwon, South Korea	Jan 15, 2025
The 4th Korean Logic Day Changwon, South Korea	Jan 13– 15, 2025
32th KIAS Combinatorics Workshop Busan, South Korea	Dec 19– 21, 2024
2024 KMS Fall Meeting Sungkyunkwan University, Suwon, South Korea	Oct 24– 26, 2024
2024 Korean Student Combinatorics Workshop (KSCW 2024) Gongju, South Korea	Jul 29– Aug 2, 2024
2024 Summer School on Combinatorics and Algorithms KIAST, Daejeon, South Korea	Jul 22– 26, 2024
Small workshop on parameterized complexity POSTECH, Pohang, South Korea	Apr 30– May 1, 2024
30th KIAS Combinatorics workshop KIAS, Seoul, South Korea	Mar 8– 9, 2024
29th KIAS Combinatorics workshop Busan, South Korea	Dec 26– 28, 2023

2023 KMS Fall Meeting Seoul National University, Seoul, South Korea	Oct 26– 28, 2023
2023 Combinatorics Workshop Yonsei University, Seoul, South Korea	Aug 16– 18, 2023
2023 KIAS Summer Workshop on Graph Theory KIAS, Seoul, South Korea	Aug 8– 9, 2023
Lecture Series on Combinatorics at Hanyang University Hanyang University, Seoul, South Korea	Aug 1– 2, 2023
28th KIAS Combinatorics workshop Jeju, South Korea	Jun 1– 3, 2023
Seminar on Combinatorial Algorithms in Hanyang University Hanyang University, Seoul, South Korea	Apr 4– 5, 2023
27th KIAS Combinatorics workshop KIAS, Seoul, South Korea	Feb 23– 24, 2023
Workshop on Graph Coloring and Related Topics Konkuk University, Seoul, South Korea	Jan 28– 29, 2023
26th KIAS Combinatorics workshop Busan, South Korea	Dec 20– 22, 2022
2022 combinatorics workshop GIST, Gwangju, South Korea	Sep 30– Oct 1, 2022
2022 School on Structural Graph Theory in Hanyang University Hanyang University, Seoul, South Korea	Sep 16– 17, 2022
Combinatorics Incheon Workshop on Jeju Jeju National University, Jeju, South Korea	Jul 25– 29, 2022
25th KIAS Combinatorics workshop Jeju, South Korea	Jun 15– 18, 2022

Workshops and Conferences (International)

GROW 2026 Będlewo, Poland	Sep 27– Oct 1, 2026
ALGO 2026 L'Aquila, Italy	Aug 31– Sep 4, 2026
ASIACOMB 2026 Daejeon, South Korea	Aug 24– 28, 2026
International Conference for the 80th Anniversary of the KMS Seoul, South Korea	Jun 22– 25, 2026
Focused Workshop on Erdős–Pósa Problems Będlewo, Poland	Mar 15– 20, 2026
SIAM Conference on Discrete Mathematics (DM24)	Jul 8– 11, 2024

Spokane, Washington, U.S.

3rd Korea-China Young Scholar's Symposium on Discrete Mathematics Jun 24– 27, 2024
Jeju, South Korea

WG 2023 – Graph-Theoretic Concepts in Computer Science Jun 28– 30, 2023
University of Fribourg, Fribourg, Switzerland

Research Visits

ENS de Lyon, Lyon, France (Host: Édouard Bonnet) Sep 6– 25, 2026
Nankai University, Tianjin, China (Host: Shenwei Huang) Sep 21– Oct 1, 2025

Academic Services

- Co-organizing 2026 Korean Student Combinatorics Workshop (KSCW 2026, 2026 조합론 학생 워크숍)

Awards

Best Paper Award. "A characterization of graphs of radius- r flip-width at most 2" Feb, 2026
Graduate Student Paper Competition, Institute of Natural Sciences, Hanyang University